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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CASE STUDY 1

TOPIC: Diabetes Data Cleaning and Analysis

P24CS904

DATA SCIENCE

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1. Abstract

Healthcare data plays an important role in understanding patient conditions, treatment patterns and hospital outcomes. However, raw healthcare datasets often contain missing values, inconsistent categories, duplicate records and unusual numerical observations.

This project focuses on cleaning and transforming the Diabetes 130-US Hospitals dataset obtained from the UCI Machine Learning Repository.

The dataset contains 101,766 hospital encounters collected from 130 US hospitals between 1999 and 2008. It includes patient demographics, admission information, medical diagnoses, laboratory procedures, medication details and hospital readmission outcomes.

The main objective is to assess data quality, handle missing values represented by question marks, identify outliers, standardise inconsistent categories, normalise numerical fields and integrate ICD-9 diagnosis descriptions.

The encounter-level dataset is subsequently transformed into a patient-level dataset containing one row per patient.

Python, Pandas and NumPy are used for data preprocessing and transformation, while Jupyter Notebook is used to document the complete workflow**.**

2. Introduction

Healthcare organisations generate large volumes of data through electronic health records, laboratory systems and hospital management systems.

Raw medical data cannot always be used directly for analysis because it may contain incomplete information, inconsistent representations and repeated patient records.

The Diabetes 130-US Hospitals dataset provides an opportunity to study these problems using practical data engineering techniques.

Each row represents a hospital encounter rather than a unique individual. A patient may have multiple hospital admissions during the observation period.

Therefore, cleaning this dataset requires two levels of analysis:

- Encounter-level data quality assessment.
- Patient-level data aggregation and transformation**.**

3. Problem Statement

The raw diabetes dataset contains missing values, inconsistent categorical values, numerical variations and multiple encounters belonging to the same patient.

These issues can reduce the reliability of healthcare analytics.

4. Dataset Description

Dataset Name: Diabetes 130-US Hospitals for Years 1999–2008.

Source: UCI Machine Learning Repository.

Dataset URL:

https://archive.ics.uci.edu/dataset/296/diabetes+130-us-hospitals+for-years+1999-2008

Dataset characteristics

Attribute	Description
Total encounters	101,766
Number of hospitals	130
Number of features	47
Observation period	1999–2008
Data type	Structured healthcare data
Main subject	Diabetes-related inpatient care
Unique patient identifier	patient_nbr
Encounter identifier	encounter_id
Outcome variable	Readmitted

The dataset includes information about:

- Patient demographics.
- Hospital admission and discharge.
- Length of hospital stay.
- Laboratory procedures.
- Medical diagnoses.
- Medication prescriptions.
- Previous hospital visits.
- Readmission status.

5. Project Objectives

1. Inspect the raw dataset and understand its structure.
2. Identify missing values and question-mark codes.
3. Detect duplicate and inconsistent records.
4. Identify numerical outliers.

5. Standardise categorical variables.
6. Normalise numerical fields.
7. Merge diagnosis information with an ICD-9 lookup table.
8. Transform encounter-level data into one row per patient.
9. Export the cleaned datasets.
10. Document every cleaning decision.

6. Initial Data Inspection

The first stage involves loading the raw dataset and understanding its dimensions, column names and data types.

Python implementation

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("diabetic_data.csv")

print("Dataset Shape:", df.shape)

print(df.head())

print(df.info())

print(df.describe(include="all"))
```

6.1 Dataset structure

The following checks are performed:

- Total number of rows.
- Total number of columns.
- Column data types.
- Duplicate records.
- Missing values.

- Unique values in categorical columns.
- Numerical ranges.

6.2 Missing value identification

In this dataset, missing information is frequently represented by the string ?.

Therefore, these values must be converted into proper missing values before performing analysis.

```
df = pd.read_csv(
   "diabetic_data.csv",
   na_values=["?"]
)

missing = df.isnull().sum()

missing_percentage = (
   df.isnull().mean() * 100
)

missing_report = pd.DataFrame({
   "Missing_Count": missing,
   "Missing_Percentage": missing_percentage
})

print(
   missing_report.sort_values(
      by="Missing_Percentage",
      ascending=False
   )
)
```

7.Numerical Data Standardisation

Numerical variables may have different measurement ranges.

For example:

- Time in hospital is measured in days.
- Number of medications represents a count.

- Number of laboratory procedures represents another count.

Normalisation helps bring numerical features to comparable scales for analytical and machine-learning applications.

Python implementation

python

```
from sklearn.preprocessing import MinMaxScaler

scale_cols = [
    "time_in_hospital",
    "num_lab_procedures",
    "num_procedures",
    "num_medications",
    "number_outpatient",
    "number_emergency",
    "number_inpatient",
    "number_diagnoses"
]

scaler = MinMaxScaler()

df_normalised = df.copy()

df_normalised[scale_cols] = (
    scaler.fit_transform(
        df[scale_cols]
    )
)

print(
    df_normalised[scale_cols].head()
)
```

8.ICD-9 Diagnosis Mapping

The dataset contains three diagnosis columns:

- diag_1 – Primary diagnosis.
- diag_2 – Secondary diagnosis.
- diag_3 – Additional secondary diagnosis.

These columns contain ICD-9 diagnosis codes.

ICD-9 codes can be connected to a lookup table containing readable diagnosis descriptions.

This improves the interpretability of the healthcare dataset.

9.Conclusion

This project demonstrates a complete data quality assessment and cleaning workflow using the Diabetes 130-US Hospitals dataset.

The analysis begins with understanding the raw dataset and identifying missing values, duplicates, inconsistent categories and numerical outliers.

Missing values represented by question marks are converted into proper null values. Categorical features are standardised, and numerical variables are examined for unusual observations.

Min-Max normalisation and Z-score standardisation are implemented to prepare numerical variables for further analytical tasks.

ICD-9 diagnosis codes are enriched using an external lookup table, making clinical diagnosis information easier to interpret.

Finally, multiple hospital encounters are consolidated into a single patient-level dataset using patient identifiers and carefully selected aggregation rules.

The project demonstrates the importance of data cleaning, data integrity, reproducibility and documentation in healthcare analytics.

10.References

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